UML Design Package

Use case, class, sequence, activity and state diagrams with cohesion and loose-coupling justification

1. Architecture Overview & Diagram Consistency

The ClubSphere UML Package models the platform across five internally consistent views: **Use Case** (functional scope across 5 actors), **Class** (structural domain model split into 4 cohesive packages), **Sequence** (time-ordered message choreography for proposing an event and getting it approved), **Activity** (cross-functional swimlanes with parallel fork/join processing), and **State Machine** (lifecycle of the **EventProposal** aggregate root and its synchronized venue lock invariants).

2. Use Case Diagram

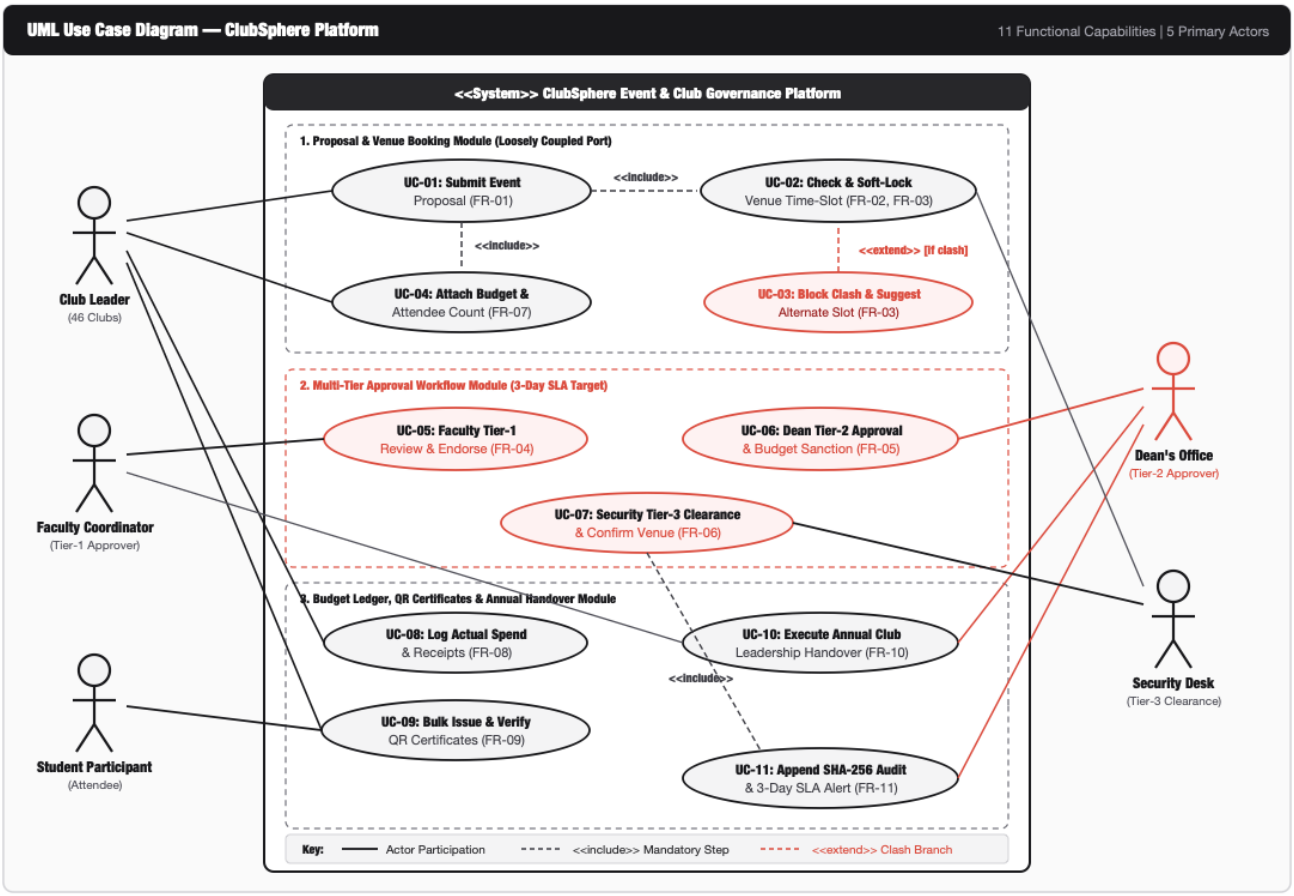


Figure 1 - UML Use Case Diagram mapping 5 campus actors to 11 functional use cases (UC-01 to UC-11).

3. Class Diagram

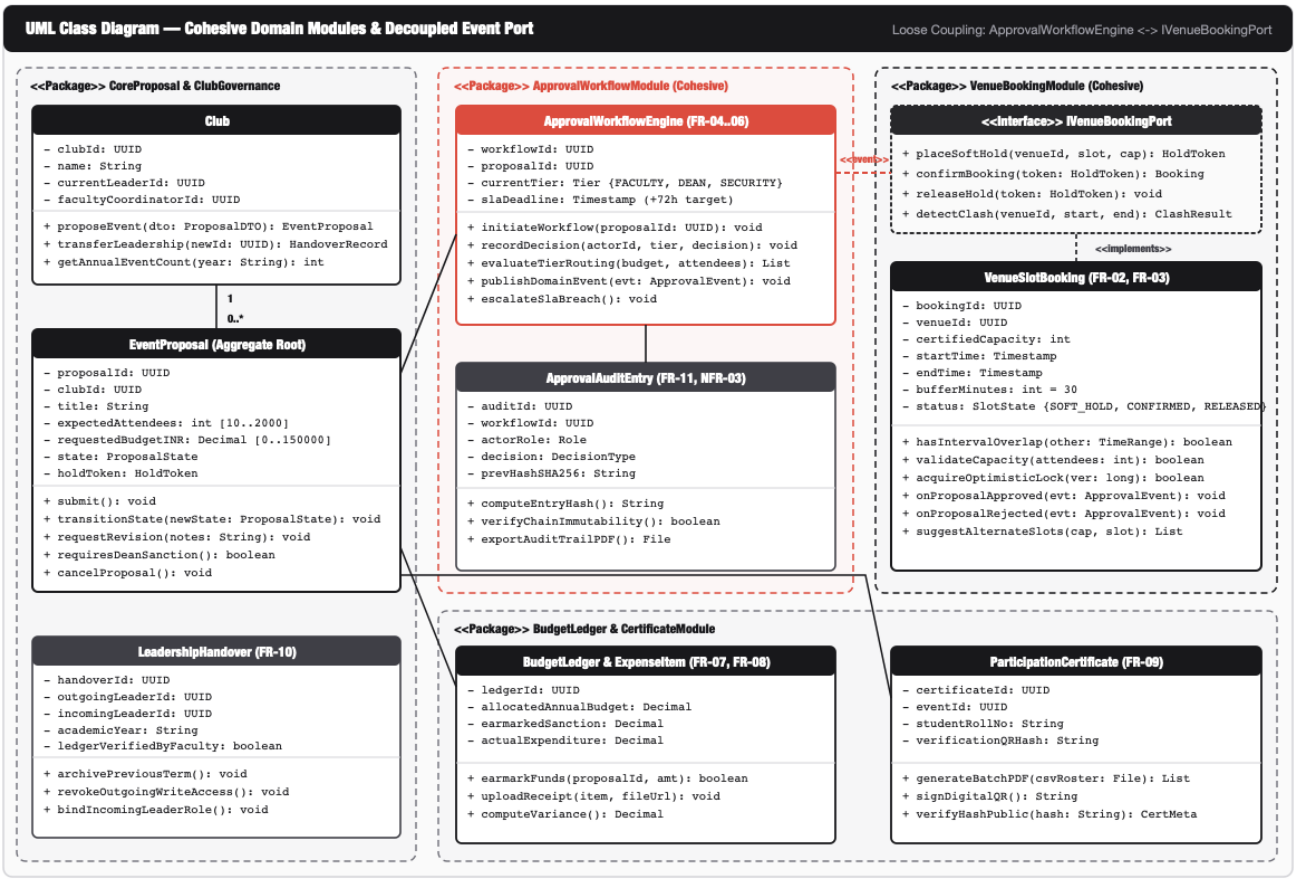


Figure 2 - UML Class Diagram showing 4 cohesive domain packages and the decoupled IVenueBookingPort contract.

4. Sequence Diagram — 'Propose an Event and Get It Approved'

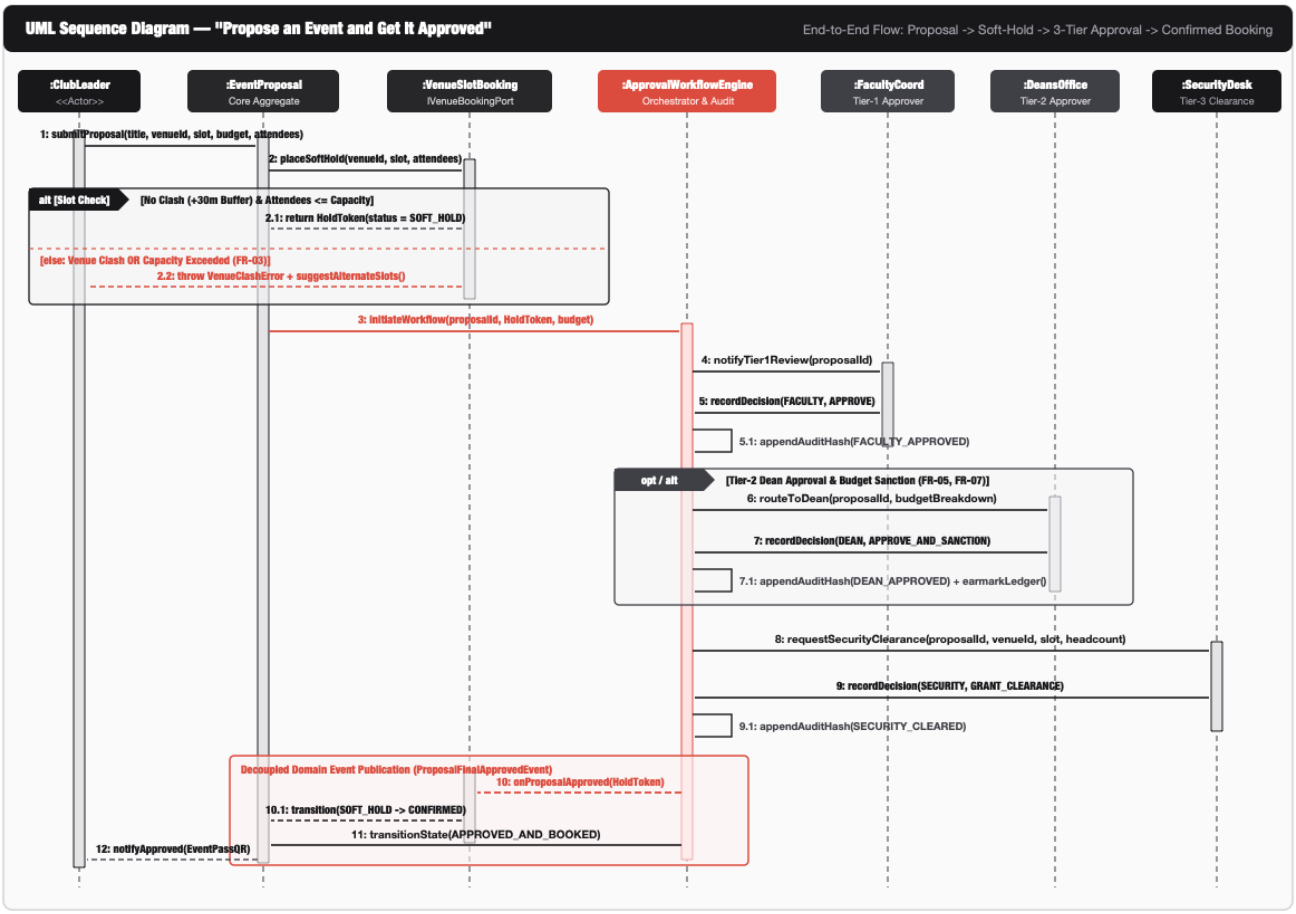


Figure 3 - End-to-end sequence diagram for proposing an event, acquiring a SOFT_HOLD, clearing 3 approval tiers, and confirming the venue.

5. Activity Diagram

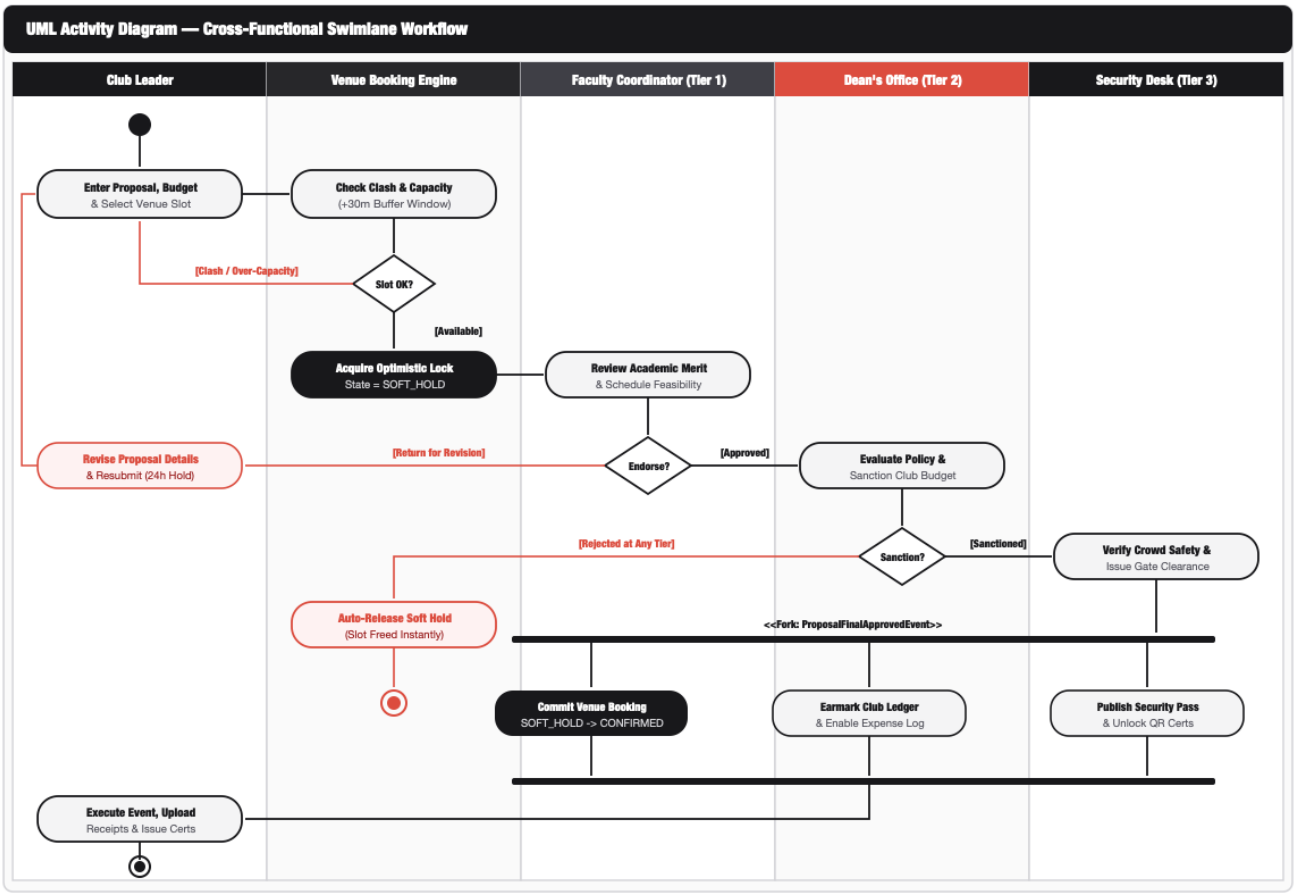


Figure 4 - Cross-functional swimlane activity diagram across Club Leader, Venue Engine, Faculty, Dean, and Security Desk.

6. State Machine Diagram — Event Proposal Lifecycle

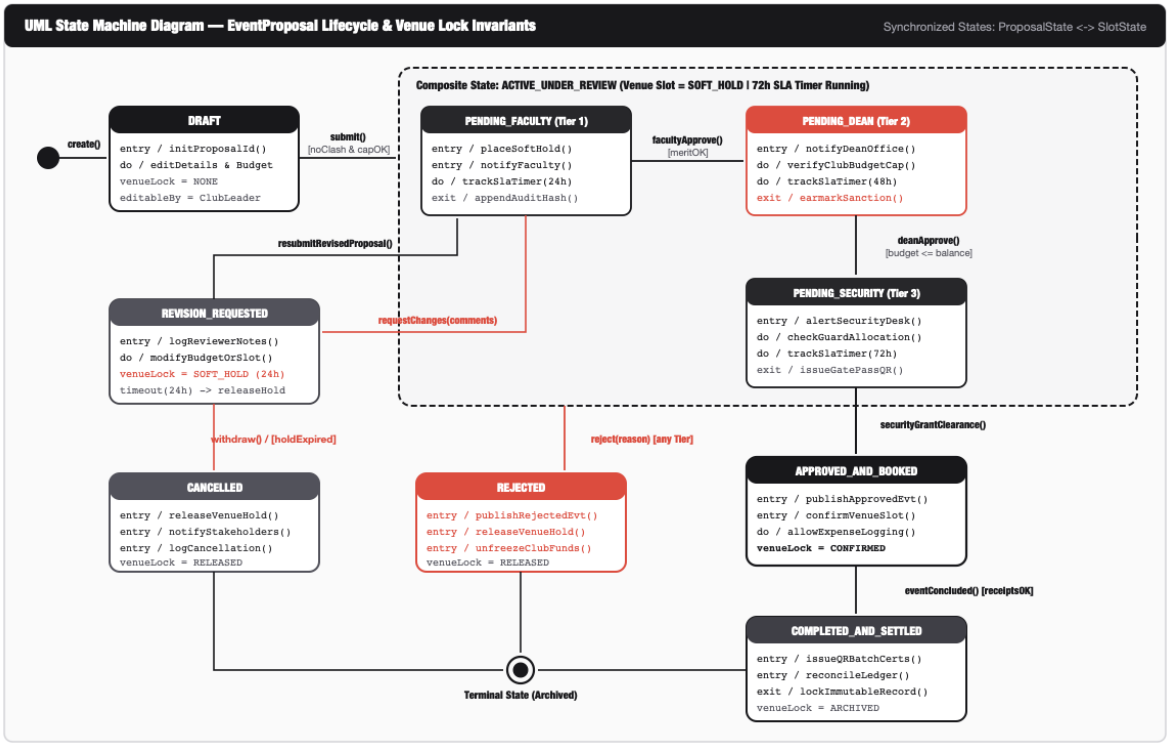


Figure 5 - UML State Machine for EventProposal showing composite review states and venue lock transitions (NONE -> SOFT_HOLD -> CONFIRMED / RELEASED).

7. Cohesion & Loose Coupling Justification (Approval vs. Venue Booking)

7.1 Why Approval Workflow and Venue Booking Must Be Decoupled

In a tightly coupled design, **ApprovalWorkflowEngine** would execute raw SQL queries against the venue calendar table during Faculty or Dean sign-off, and **VenueSlotBooking** would inspect approval hierarchy tables. That coupling creates three critical risks: (1) policy changes to approval tiers (e.g., adjusting the Rs. 0 fast-track threshold) would break calendar clash logic; (2) holding database locks across multi-day human approvals would cause contention; and (3) junior student maintainers taking over at Month 9 would face entangled business rules.

7.2 Three Mechanisms Enforcing Loose Coupling

- **1. Dependency Inversion via IVenueBookingPort & Opaque HoldToken (Stamp Coupling Only):** When a proposal is submitted, **EventProposal** calls **IVenueBookingPort.placeSoftHold(venueId, slot, attendees)**. The venue module checks capacity and the ±30-minute buffer and returns an opaque **HoldToken**. **ApprovalWorkflowEngine** stores only this token ID and has zero knowledge of room capacities, building layouts, or interval overlap math.
- **2. Asynchronous Domain Event Choreography (Pub/Sub):** When Tier-3 Security grants final clearance (or when any tier rejects the proposal), **ApprovalWorkflowEngine** publishes an immutable domain event (**ProposalFinalApprovedEvent** or **ProposalRejectedEvent**). **VenueSlotBooking** subscribes to these events and atomically transitions its slot from **SOFT_HOLD** to **CONFIRMED** or **RELEASED**.
- **3. Two-Phase Reservation State (SOFT_HOLD -> CONFIRMED):** Temporal clash prevention is enforced via a PostgreSQL GiST exclusion constraint on persisted slot states rather than long-lived database row locks.

7.3 High Functional Cohesion Across Modules & Cross-Diagram Consistency

Module / Package	Single Cohesive Responsibility	Encapsulated State & Rules	Cross-Diagram Trace
ApprovalWorkflowEngine	Orchestrates 3-tier routing (Faculty -> Dean -> Security), 72h SLA timers, and SHA-256 audit hash chaining	workflowId, currentTier, slaDeadline, ApprovalAuditEntry chain	UC-05..07, UC-11; Seq Steps 3–9; State Composite Review
VenueBookingService	Manages physical venue capacities, ±30m buffer interval overlap math, optimistic locking, and alternate slot suggestions	venueId, certifiedCapacity, startTime, endTime, bufferMinutes=30, SlotState	UC-02, UC-03; Seq Steps 2 & 10; Act Lane 2
BudgetLedger	Tracks 46 club annual allocations, earmarks sanctioned funds, and enforces the 7-day post-event receipt gate	allocatedAnnualBudget, earmarkedSanction, actualExpenditure	UC-04, UC-08; Seq Step 7.1; Act Fork Branch B
Certificate & Handover	Generates batch SHA-256 QR PDF certificates and executes annual club leadership handover	certificateId, verificationQRHash, handoverId, academicYear	UC-09, UC-10; Class Packages 1 & 4